DATABASE MANAGEMENT SYSTEMS
[ENCS204]

Bachelor of Technology
in
Computer Science and Engineering

Submitted by
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CAPSTONE ASSIGNMENT

Cybersecurity Incident Response System

INTRODUCTION

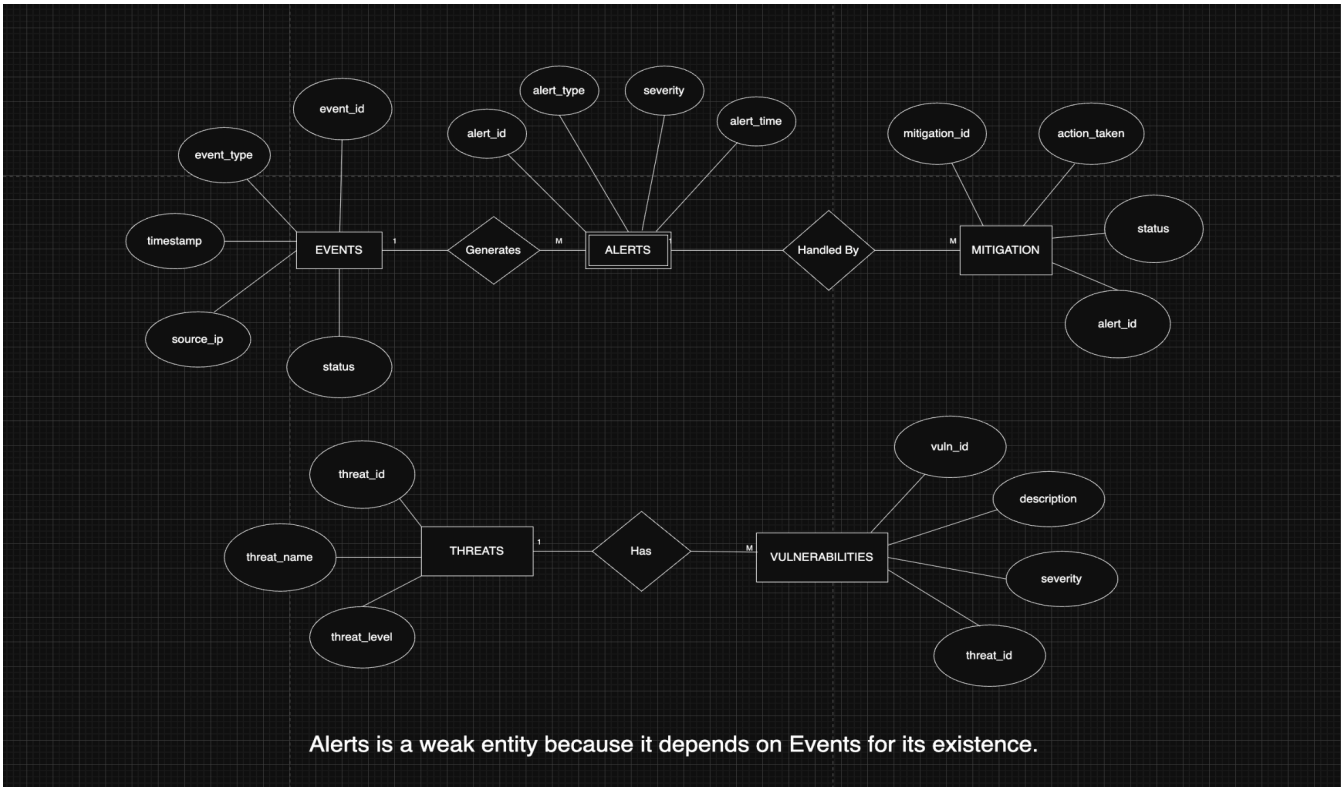
A Cybersecurity Incident Response System is used to detect, analyze, and respond to security incidents in an organization.
This system helps in identifying threats, vulnerabilities, and taking appropriate mitigation actions.
In this project, a database is designed to manage cybersecurity events, alerts, threats, vulnerabilities, and mitigation actions.
SQL queries are used to analyze data, identify attack patterns, and prioritize response actions.

DATABASE TABLES

The following tables are created in the database:

1. Events
2. Alerts
3. Threats
4. Vulnerabilities
5. Mitigation

ER DIAGRAM



SQL QUERIES:

STEP 1:

CREATE DATABASE CyberSecurityDB;

1	19:09:14	CREATE DATABASE CyberSecurityDB	1 row(s) affected	0.018 sec

STEP 2:

USE CyberSecurityDB;

2	19:10:50	USE CyberSecurityDB	0 row(s) affected	0.029 sec

STEP 3:

```
CREATE TABLE Events (
    event_id INT PRIMARY KEY,
    event_type VARCHAR(50),
    timestamp DATETIME,
    source_ip VARCHAR(20),
    status VARCHAR(20)
);
```

3	19:17:37	CREATE TABLE Events (event_id INT PRIMA...	0 row(s) affected	0.048 sec

STEP 4:

```
CREATE TABLE Alerts (
    alert_id INT PRIMARY KEY,
    event_id INT,
    alert_type VARCHAR(50),
    severity VARCHAR(20),
    alert_time DATETIME,
    FOREIGN KEY (event_id) REFERENCES
Events(event_id)
);
```

5	19:18:50	CREATE TABLE Alerts (alert_id INT PRIMARY...	0 row(s) affected	0.023 sec

STEP 5:

```
CREATE TABLE Threats (
    threat_id INT PRIMARY KEY,
    threat_name VARCHAR(50),
    threat_level VARCHAR(20)
);
```

6	19:20:30	CREATE TABLE Threats (threat_id INT PRIM...	0 row(s) affected	0.023 sec

STEP 6:

```
CREATE TABLE Vulnerabilities (
    vuln_id INT PRIMARY KEY,
    description VARCHAR(100),
    severity VARCHAR(20),
    threat_id INT,
    FOREIGN KEY (threat_id) REFERENCES
Threats(threat_id)
);
```

7	19:21:15	CREATE TABLE Vulnerabilities (	vuln_id INT P... 0 row(s) affected	0.014 sec

STEP 7:

```
CREATE TABLE Mitigation (
    mitigation_id INT PRIMARY KEY,
    alert_id INT,
    action_taken VARCHAR(100),
    status VARCHAR(20),
    FOREIGN KEY (alert_id) REFERENCES Alerts(alert_id)
);
```

8	19:22:24	CREATE TABLE Mitigation (	mitigation_id INT... 0 row(s) affected	0.020 sec

STEP 8:

```
INSERT INTO Events VALUES
(1, 'Login Attempt', '2026-03-01 10:00:00', '192.168.1.1',
'Failed'),
(2, 'File Access', '2026-03-01 11:00:00', '192.168.1.2',
'Success'),
(3, 'Malware Detection', '2026-03-01 12:00:00', '192.168.1.3',
'Blocked'),
(4, 'Unauthorized Access', '2026-03-01 01:00:00',
'192.168.1.4', 'Failed');

SELECT * FROM Events;
```

100% 22:1

Result Grid Filter Rows: Search Edit: Export/Import: Result Grid

event_id	event_type	timestamp	source_ip	status
1	Login Attempt	2026-03-01 10:00:00	192.168.1.1	Failed
2	File Access	2026-03-01 11:00:00	192.168.1.2	Success
3	Malware Detection	2026-03-01 12:00:00	192.168.1.3	Blocked
4	Unauthorized Access	2026-03-01 01:00:00	192.168.1.4	Failed
NULL	NULL	NULL	NULL	NULL

Events 2 Apply

Action Output

	Time	Action	Response	Duration / Fetch Time
6	19:20:30	CREATE TABLE Threats (threat_id INT PRIM...	0 row(s) affected	0.023 sec
7	19:21:15	CREATE TABLE Vulnerabilities (vuln_id INT P...	0 row(s) affected	0.014 sec
8	19:22:24	CREATE TABLE Mitigation (mitigation_id INT...	0 row(s) affected	0.020 sec
9	19:23:53	INSERT INTO Events VALUES (1, 'Login Attempt...	4 row(s) affected Records: 4 Duplicates: 0 Warnings...	0.013 sec
10	19:24:17	SELECT * FROM Events LIMIT 0, 1000	4 row(s) returned	0.00047 sec / 0.0000...

STEP 9:

```
INSERT INTO Alerts VALUES
(101, 1, 'Brute Force', 'High', '2026-03-01 10:05:00'),
(102, 3, 'Virus Alert', 'Critical', '2026-03-01 12:05:00'),
(103, 4, 'Intrusion', 'High', '2026-03-01 01:10:00');

SELECT * FROM Alerts;
```

100% 22:1

Result Grid Filter Rows: Search Edit: Export/Import: Result Grid

alert_id	event_id	alert_type	severity	alert_time
101	1	Brute Force	High	2026-03-01 10:05:00
102	3	Virus Alert	Critical	2026-03-01 12:05:00
103	4	Intrusion	High	2026-03-01 01:10:00
NULL	NULL	NULL	NULL	NULL

Alerts 3 Apply

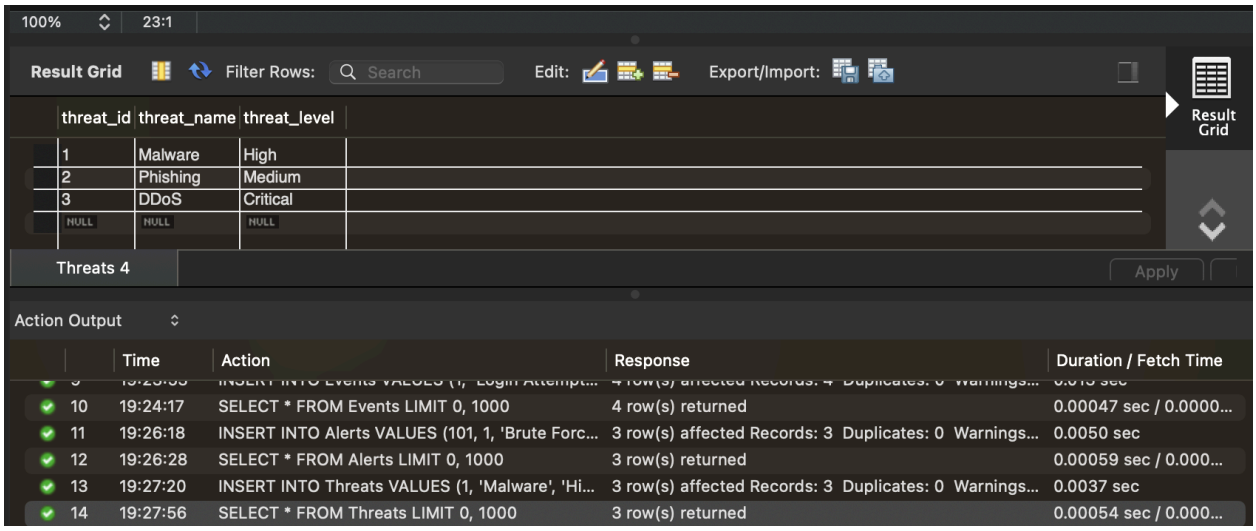
Action Output

	Time	Action	Response	Duration / Fetch Time
8	19:22:24	CREATE TABLE Mitigation (mitigation_id INT...	0 row(s) affected	0.020 sec
9	19:23:53	INSERT INTO Events VALUES (1, 'Login Attempt...	4 row(s) affected Records: 4 Duplicates: 0 Warnings...	0.013 sec
10	19:24:17	SELECT * FROM Events LIMIT 0, 1000	4 row(s) returned	0.00047 sec / 0.0000...
11	19:26:18	INSERT INTO Alerts VALUES (101, 1, 'Brute Forc...	3 row(s) affected Records: 3 Duplicates: 0 Warnings...	0.0050 sec
12	19:26:28	SELECT * FROM Alerts LIMIT 0, 1000	3 row(s) returned	0.00059 sec / 0.000...

STEP 10:

INSERT INTO Threats VALUES
(1, 'Malware', 'High'),
(2, 'Phishing', 'Medium'),
(3, 'DDoS', 'Critical');

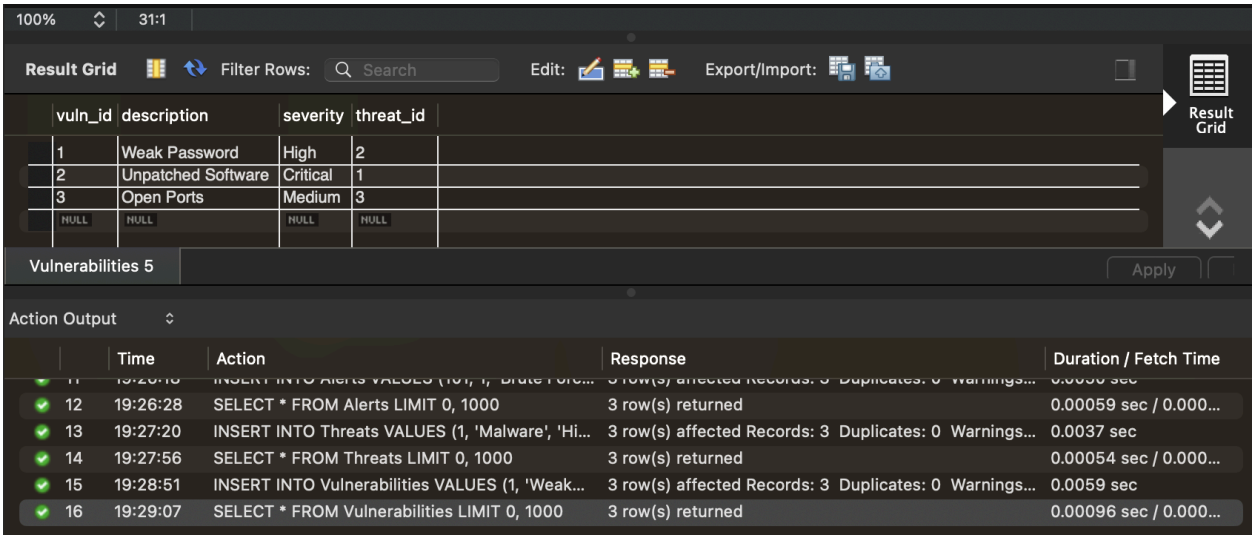
SELECT * FROM Threats;



STEP 11:

INSERT INTO Vulnerabilities VALUES
(1, 'Weak Password', 'High', 2),
(2, 'Unpatched Software', 'Critical', 1),
(3, 'Open Ports', 'Medium', 3);

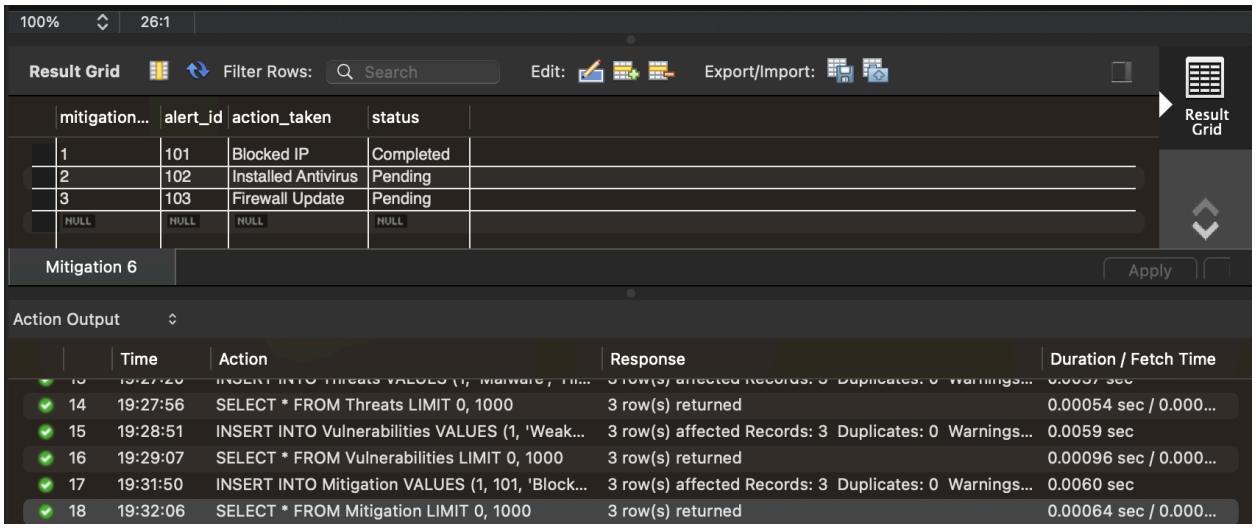
SELECT * FROM Vulnerabilities;



STEP 12:

```
INSERT INTO Mitigation VALUES
(1, 101, 'Blocked IP', 'Completed'),
(2, 102, 'Installed Antivirus', 'Pending'),
(3, 103, 'Firewall Update', 'Pending');

SELECT * FROM Mitigation;
```

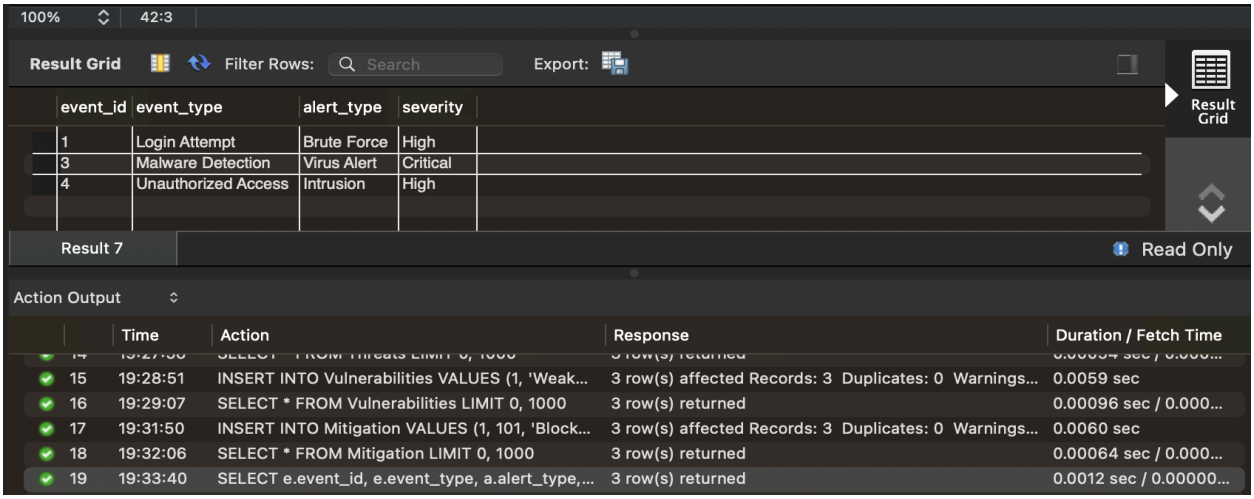


mitigation...	alert_id	action_taken	status
1	101	Blocked IP	Completed
2	102	Installed Antivirus	Pending
3	103	Firewall Update	Pending
NULL	NULL	NULL	NULL

	Time	Action	Response	Duration / Fetch Time
14	19:27:56	SELECT * FROM Threats LIMIT 0, 1000	3 row(s) returned	0.00054 sec / 0.000...
15	19:28:51	INSERT INTO Vulnerabilities VALUES (1, 'Weak...	3 row(s) affected Records: 3 Duplicates: 0 Warnings...	0.0059 sec
16	19:29:07	SELECT * FROM Vulnerabilities LIMIT 0, 1000	3 row(s) returned	0.00096 sec / 0.000...
17	19:31:50	INSERT INTO Mitigation VALUES (1, 101, 'Block...	3 row(s) affected Records: 3 Duplicates: 0 Warnings...	0.0060 sec
18	19:32:06	SELECT * FROM Mitigation LIMIT 0, 1000	3 row(s) returned	0.00064 sec / 0.000...

STEP 13:

```
SELECT e.event_id, e.event_type, a.alert_type, a.severity
FROM Events e
JOIN Alerts a ON e.event_id = a.event_id;
```

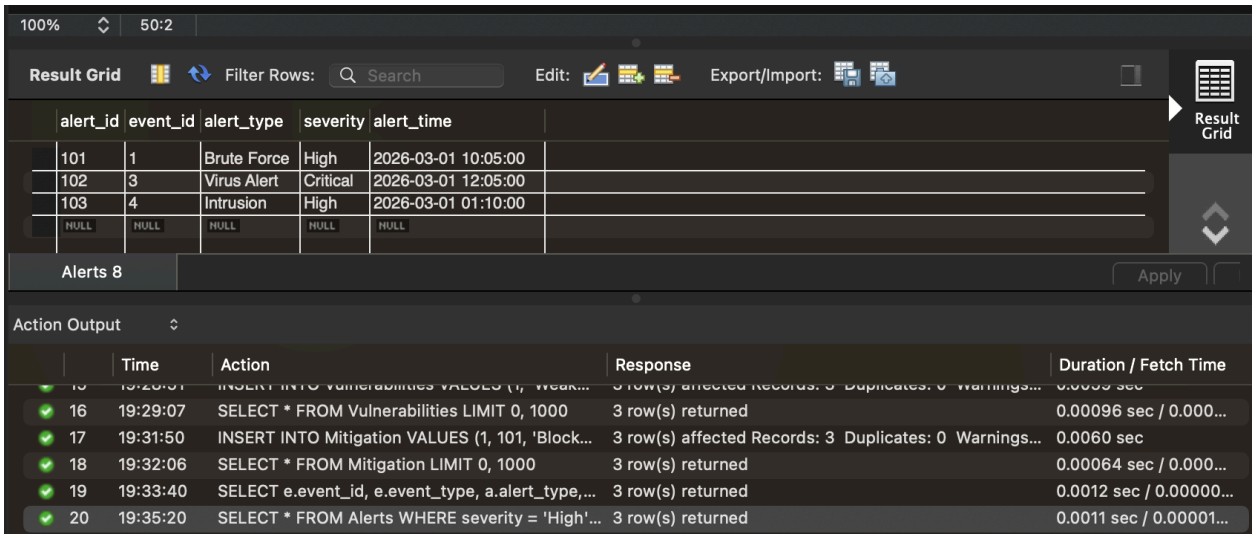


event_id	event_type	alert_type	severity
1	Login Attempt	Brute Force	High
3	Malware Detection	Virus Alert	Critical
4	Unauthorized Access	Intrusion	High

	Time	Action	Response	Duration / Fetch Time
15	19:28:51	INSERT INTO Vulnerabilities VALUES (1, 'Weak...	3 row(s) affected Records: 3 Duplicates: 0 Warnings...	0.0059 sec
16	19:29:07	SELECT * FROM Vulnerabilities LIMIT 0, 1000	3 row(s) returned	0.00096 sec / 0.000...
17	19:31:50	INSERT INTO Mitigation VALUES (1, 101, 'Block...	3 row(s) affected Records: 3 Duplicates: 0 Warnings...	0.0060 sec
18	19:32:06	SELECT * FROM Mitigation LIMIT 0, 1000	3 row(s) returned	0.00064 sec / 0.000...
19	19:33:40	SELECT e.event_id, e.event_type, a.alert_type,...	3 row(s) returned	0.0012 sec / 0.00000...

STEP 14:

```
SELECT * FROM Alerts
WHERE severity = 'High' OR severity = 'Critical';
```

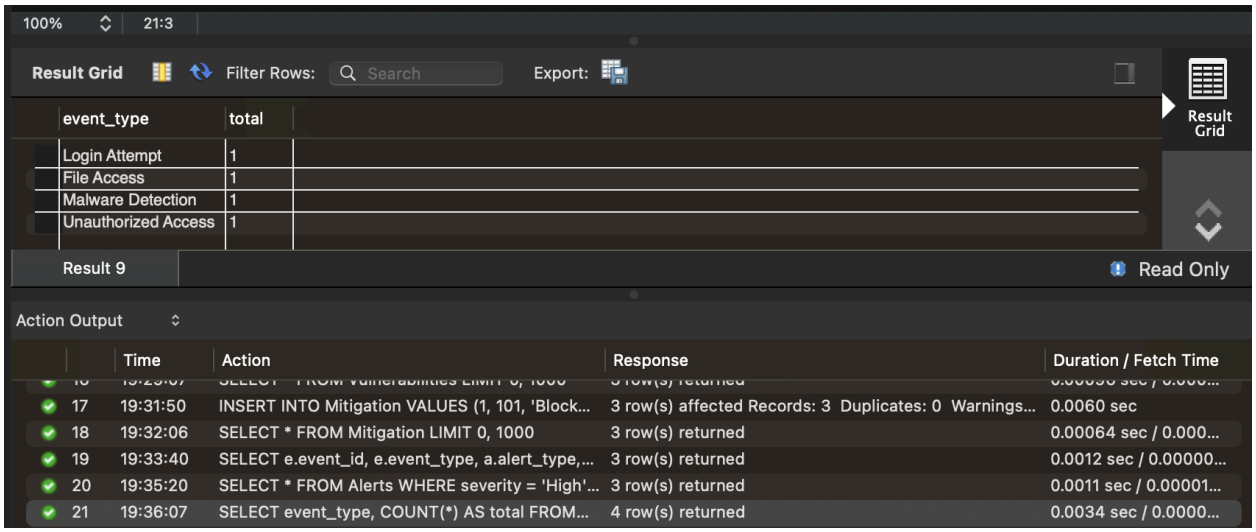


alert_id	event_id	alert_type	severity	alert_time
101	1	Brute Force	High	2026-03-01 10:05:00
102	3	Virus Alert	Critical	2026-03-01 12:05:00
103	4	Intrusion	High	2026-03-01 01:10:00
NULL	NULL	NULL	NULL	NULL

	Time	Action	Response	Duration / Fetch Time
16	19:29:07	SELECT * FROM Vulnerabilities LIMIT 0, 1000	3 row(s) returned	0.00096 sec / 0.000...
17	19:31:50	INSERT INTO Mitigation VALUES (1, 101, 'Block...	3 row(s) affected Records: 3 Duplicates: 0 Warnings...	0.0060 sec
18	19:32:06	SELECT * FROM Mitigation LIMIT 0, 1000	3 row(s) returned	0.00064 sec / 0.000...
19	19:33:40	SELECT e.event_id, e.event_type, a.alert_type,...	3 row(s) returned	0.0012 sec / 0.00000...
20	19:35:20	SELECT * FROM Alerts WHERE severity = 'High'...	3 row(s) returned	0.0011 sec / 0.00001...

STEP 15:

```
SELECT event_type, COUNT(*) AS total
FROM Events
GROUP BY event_type;
```

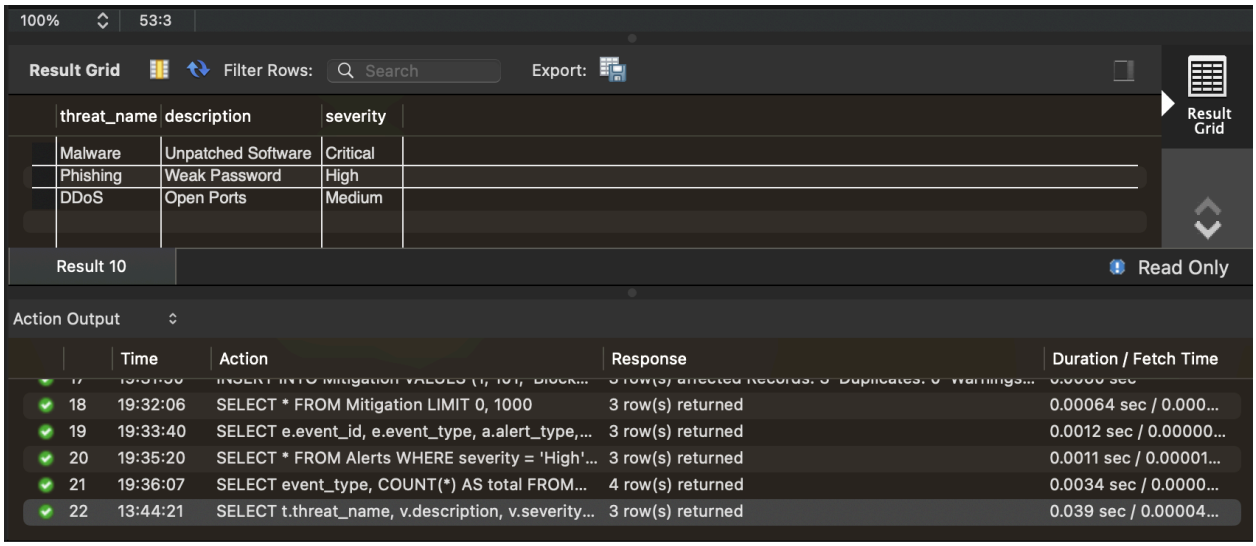


event_type	total
Login Attempt	1
File Access	1
Malware Detection	1
Unauthorized Access	1

	Time	Action	Response	Duration / Fetch Time
17	19:31:50	INSERT INTO Mitigation VALUES (1, 101, 'Block...	3 row(s) affected Records: 3 Duplicates: 0 Warnings...	0.0060 sec
18	19:32:06	SELECT * FROM Mitigation LIMIT 0, 1000	3 row(s) returned	0.00064 sec / 0.000...
19	19:33:40	SELECT e.event_id, e.event_type, a.alert_type,...	3 row(s) returned	0.0012 sec / 0.00000...
20	19:35:20	SELECT * FROM Alerts WHERE severity = 'High'...	3 row(s) returned	0.0011 sec / 0.00001...
21	19:36:07	SELECT event_type, COUNT(*) AS total FROM...	4 row(s) returned	0.0034 sec / 0.0000...

STEP 16:

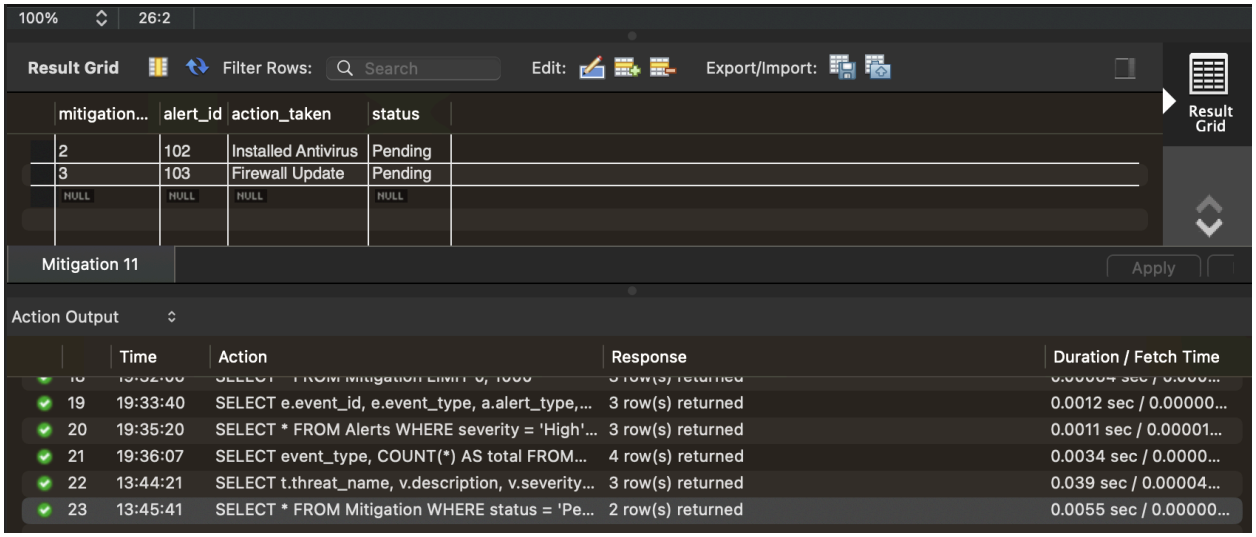
```
SELECT t.threat_name, v.description, v.severity
FROM Threats t
JOIN Vulnerabilities v ON t.threat_id = v.threat_id;
```



threat_name	description	severity
Malware	Unpatched Software	Critical
Phishing	Weak Password	High
DDoS	Open Ports	Medium

STEP 17:

```
SELECT * FROM Mitigation
WHERE status = 'Pending';
```



mitigation...	alert_id	action_taken	status
2	102	Installed Antivirus	Pending
3	103	Firewall Update	Pending
NULL	NULL	NULL	NULL

WEAK ENTITY

Alerts is a weak entity because it depends on the Events entity for its existence.
It cannot exist without an associated event and uses event_id as a foreign key.

CONCLUSION

This project demonstrates the design and implementation of a Cybersecurity Incident Response System using a relational database.
It helps in understanding how security events are managed, analyzed, and mitigated using SQL queries.